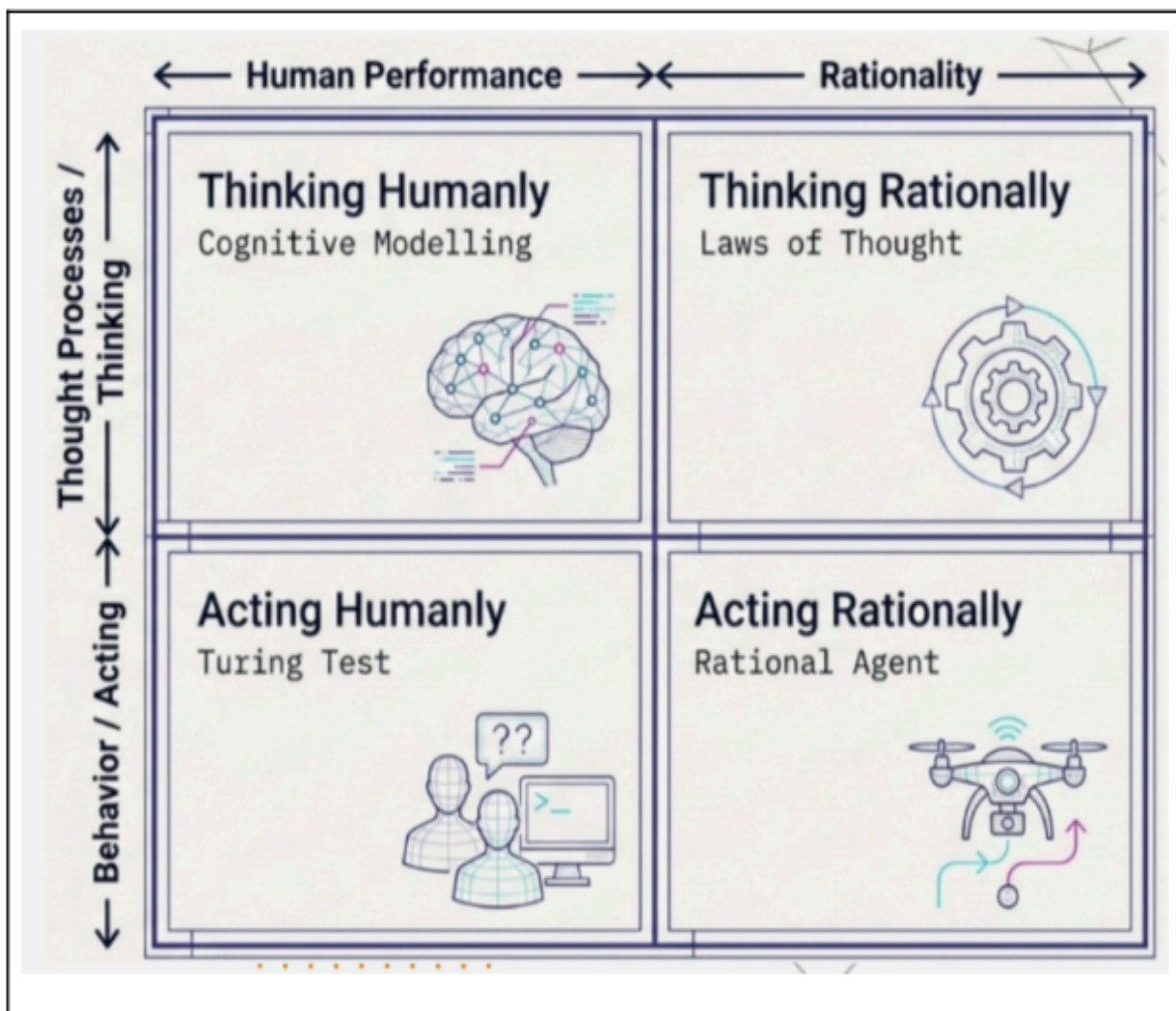


Question 1

- a) Artificial Intelligence (AI) refers to the simulation of human intelligence in machines are programmed to think, learn, and make decisions like humans. Most definitions of AI can be categorised along two key dimensions: 1) human performance versus rationality and 2) thought processes and reasoning (thinking) versus behaviour (action).
- (i) Draw a table with appropriate labels to illustrate the classification of **FOUR (4)** approaches mentioned above. (6 marks)



- (ii) Define the Turing Test approach and explain how it works. (4 marks)

- The Turing Test, proposed by Alan Turing in 1950 evaluates a computer's ability to

exhibit intelligent behavior equivalent to that of a human.

- The test involves a computer, a human respondent, and a human interrogator, with direct physical interaction avoided.
- The human interrogator with communicate with both the machines and human respondent by asking the same questions.
- The computer passes the test if the human interrogator, after posing written questions, cannot determine whether the responses come from a person or a computer

(iii) Identify another name for the Turing Test.

(1 mark)

Imitation game

- b) A task environment represents the problems that a rational agent is designed to solve. Therefore, the first step in designing an agent must always be to specify the task environment as fully as possible.

(i) Describe the **FOUR (4)** components of the **PEAS** framework.

(8 marks)

Performance Measure – This defines how the agent's success is evaluated. It includes the criteria or goals that determine whether the agent is performing well.

Environment – The surrounding in which the agent operates. It includes everything the agent interacts with while performing its tasks.

Actuators – The components that allow the agent to take action and interact with its environment.

Sensors – The input devices that allow the agent to perceive its environment by collecting data.

- (ii) Illustrate how the **PEAS** framework can be applied to describe various task environments for an autonomous delivery drone, as shown in **Figure 1**. Provide **ONE (1)** example for each of the components. (4 marks)



Figure 1. Autonomous Drone for Delivery

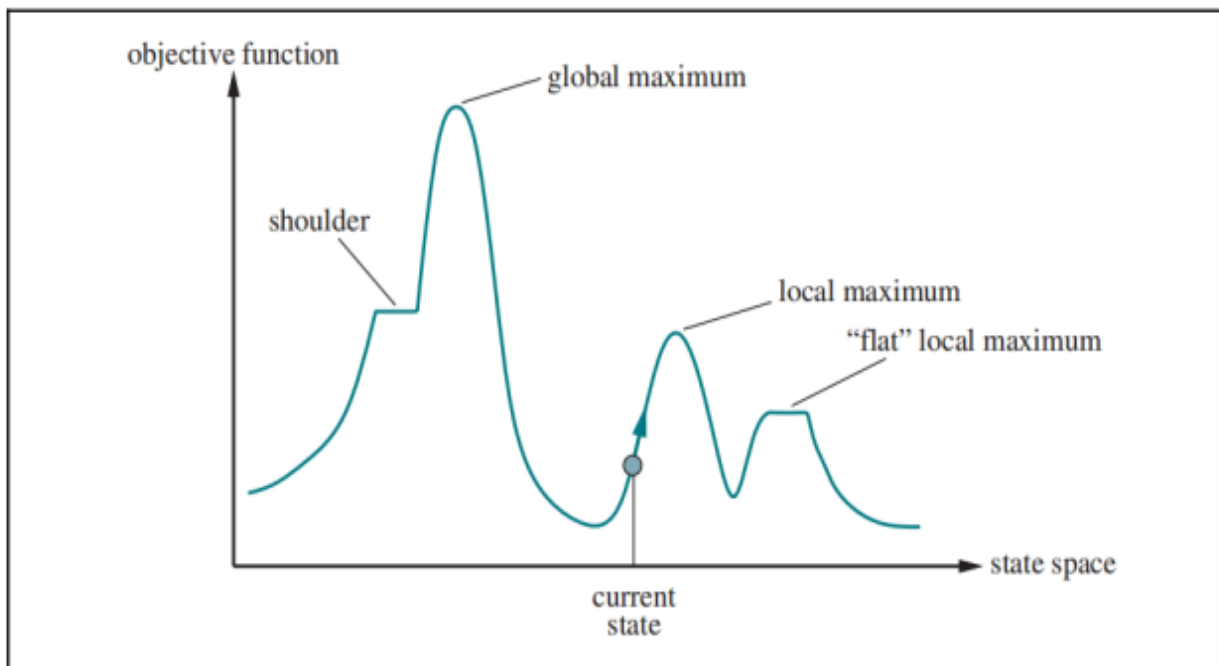
Performance Measure	Timely delivery, package safety, energy efficiency, and avoidance of collisions
Environment	Airspace, weather conditions, buildings, trees, and other drones or aircraft
Actuators	GPS navigation system, Propellers, package release mechanism
Sensors	Cameras, GPS, altimeters, obstacle detection sensors, and wind sensors.

- (iii) State whether you agree that the autonomous delivery drone has omniscience. Justify your answer. (2 marks)

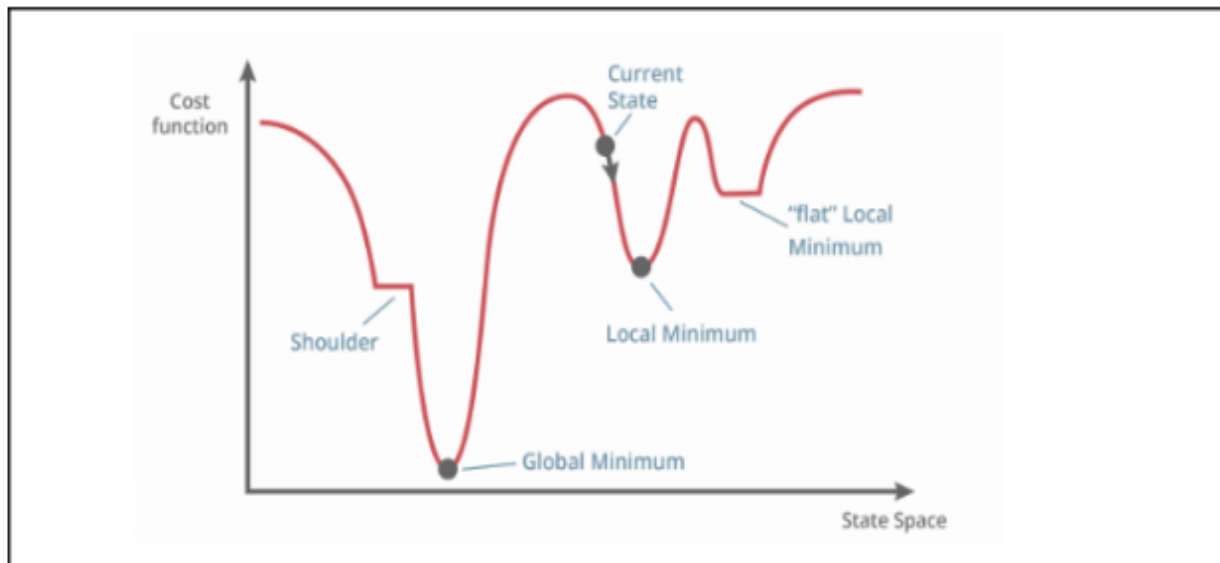
No, It doesn't know the actual outcome of its action.

Question 2

- a) **Hill climbing** is an optimisation algorithm used for solving problems where the goal is to find the best solution by iteratively making small changes. It is commonly used in AI and machine learning for local search optimisation.
- (i) Draw a state-space landscape in which its elevation corresponds to an objective function. (7 marks)



- (ii) Draw a state-space landscape in which its elevation corresponds to a cost function. (7 marks)



- (iii) Describe the **FIVE (5)** labels (current state, global maximum, local maximum, flat local maximum, shoulder) as in **Question 2 a) (i)**. Then, examine the impact of these labels on the efficiency and effectiveness of the hill-climbing algorithm. (10 marks)

Current State

- The state where the agent is **currently** located at.
- **Impact:** A poor initial state may lead to getting stuck in local maxima or plateaus, reducing efficiency.

Global Maximum

- The best possible state with the **highest value** of objective function.
- **Impact:** If the algorithm can reach this point, it ensures the best possible solution.

Local Maximum

- A state whose value is **higher** than all of its neighboring states, but **not the best one**.
- **Impact:** Reduces effectiveness, as the algorithm can converge prematurely, missing the global maximum.

Flat Local Maximum (Plateau)

- A region where the current state and its neighboring states have the **same objective-function value**.
- **Impact:** The algorithm may waste time exploring similar states or terminate without improvement.

Shoulder

- A plateau whose edge is **stretching upwards** (going in the same overall direction).
- **Impact:** The algorithm might stop early, thinking no improvement is possible, reducing efficiency.

- (iv) Identify **ONE (1)** algorithm that can improve the efficiency and effectiveness of the hill-climbing algorithm. (1 mark)

[Total: 25 marks]

Simulated Annealing

Question 3

- a) A software engineering team has developed an AI-powered map navigation application designed to find the shortest path from Station A to Station H. **Figure 4** presents a directed graph consisting of eight stations, where the values within the graph indicate the distance (in kilometers) between each station. **Table 1** displays the heuristic cost of the stations relative to Station H.

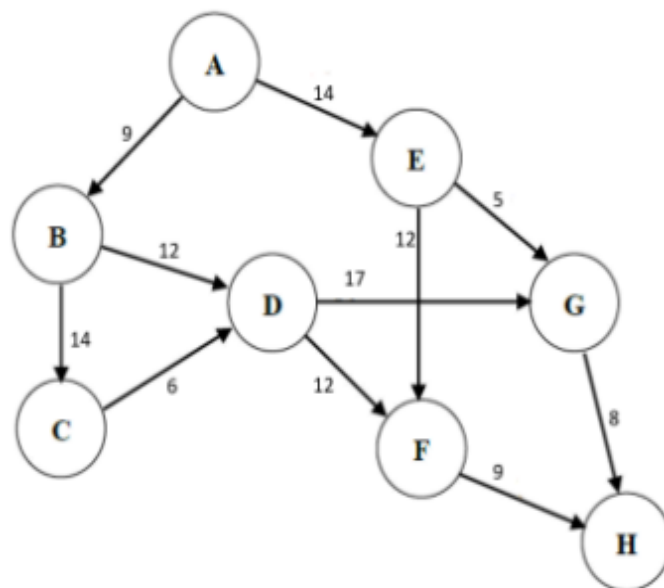


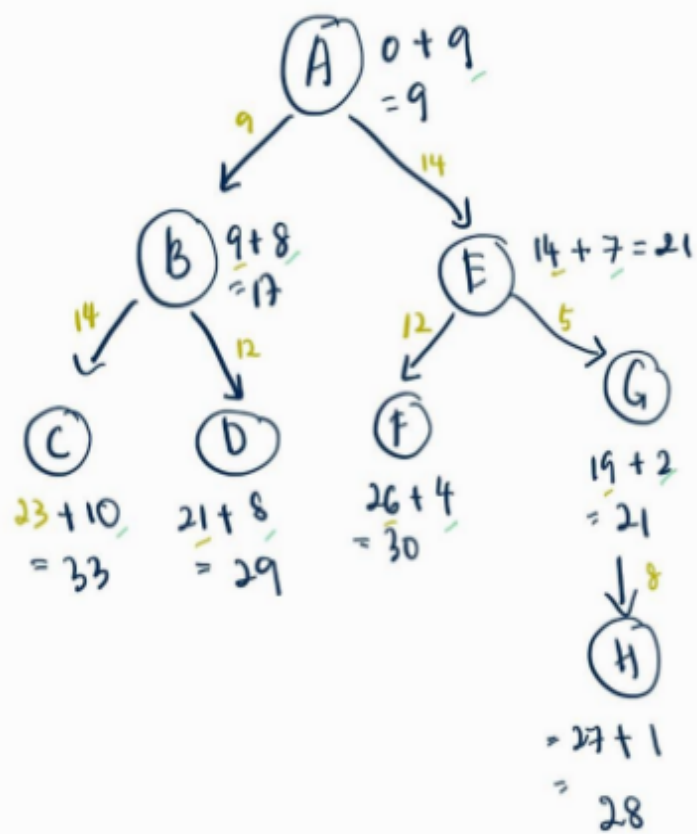
Figure 4. Search Graph

Table 1: Heuristic cost of each station

Station	A	B	C	D	E	F	G	H
Euclidean distance	9	8	10	8	7	4	2	1

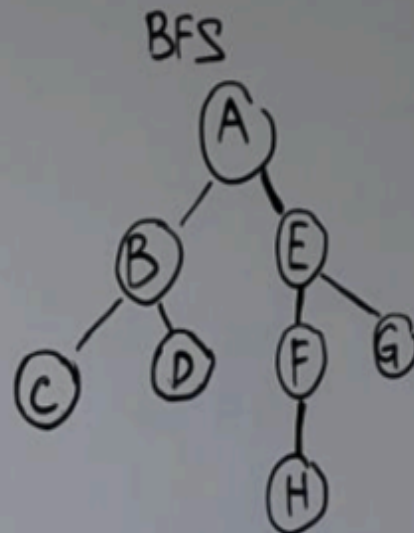
- Analyse the state space, initial state, goal state, action, transition model, and the cost function for the problem above. (6 marks)
- Based on the information provided in **Figure 4** and **Table 1**, illustrate the resulting search tree for A* search in alphabetical order. (8 marks)
- Identify the solution returned by A* search in **Question 3 a) (ii)**. (1 mark)

A*



BFS

Expanded Node	Frontier
	A
A	B, E
B	E, C, D
E	C, D, F, G
C	D, F, G
D	F, G
F	G, H

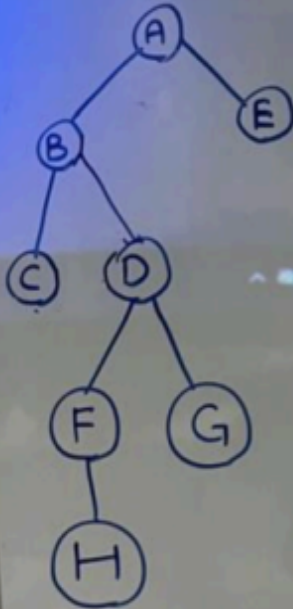


Solution Path:
A, E, F, H

Total path cost:
 $14 + 12 + 9 = 35$

DFS

DFS



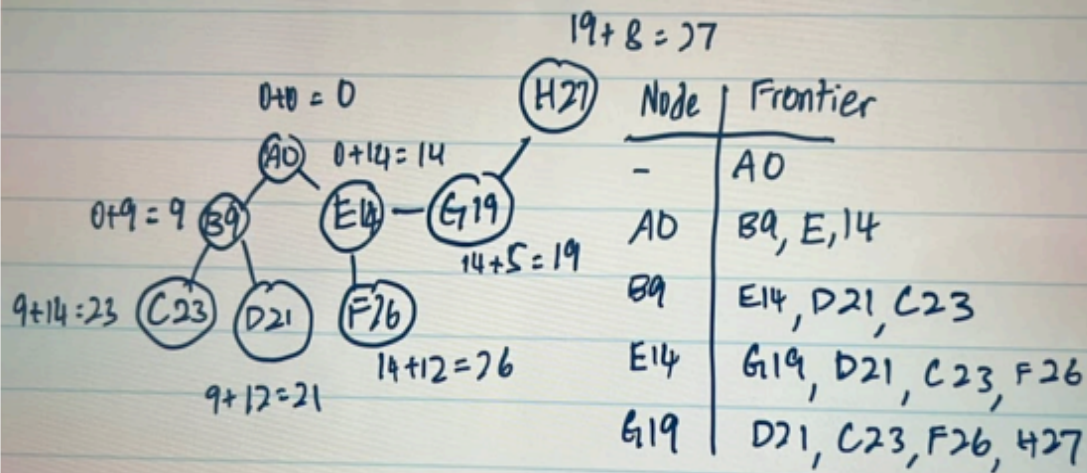
Expanded node	from
	A
A	B, E
B	C, D, E
C	D, E
D	F, G, E
F	H, G, E
H	G, E

$$\begin{array}{r} 18 \\ 24 \\ \hline 42 \end{array}$$

① solution path = A, B, D, F, H

② Total path cost = $9 + 12 + 12 + 9$
= 42

③ traversal path = A, B, C, D, F, H



Traversal path = A, B, E, G

Solution path = A, E, G, H

Total cost path = $14 + 5 + 8$
 $= 27$

Question 4

b) A university needs to schedule classes for four different courses:

1. Mathematics (M)
2. Computer Science (CS)
3. Physics (P)
4. Artificial Intelligence (AI)

The university has three available time slots and three classrooms. Each course must be assigned to a time slot and a room while satisfying several constraints. Mathematics must be scheduled before Computer Science.

Time Slots:

T1 (8:00 AM - 10:00 AM)

T2 (10:00 AM - 12:00 PM)

T3 (2:00 PM - 4:00 PM)

Classrooms:

R1, R2, R3

Lecturer Availability:

Prof. A (M) : Available at T1 or T2

Prof. B (CS) : Available at T2 or T3

Prof. C (P) : Available at T1 or T3

Prof. D (AI) : Available at T2 or T3

(i) Define a Constraint Satisfaction Problem (CSP).

(3 marks)

- A CSP is a problem of **finding values for variables that satisfy all given constraints** , where **each variable must take a value from its defined domain** . When all constraints between **variables are satisfied** , a solution is found.
- It consists of three main components: **1) variables, 2) domains, and 3) constraints.**

(ii) Analyse the case study in **Question 3 b)** as CSP by identifying its variables and each variable's domains. (5 marks)

Variables:

Course Assignments (C) : Assign each course to a classroom and a time

slot.

- $C_m = (\text{Room}, \text{Time})$ for Mathematics
- $C_{cs} = (\text{Room}, \text{Time})$ for Computer Science
- $C_p = (\text{Room}, \text{Time})$ for Physics
- $C_{ai} = (\text{Room}, \text{Time})$ for Artificial Intelligence

Domain for each variable:

Domain (C_x) = $\{(R1, T1), (R1, T2), (R1, T3), (R2, T1), (R2, T2), (R2, T3), (R3, T1), (R3, T2), (R3, T3)\}$

Constraint:

- $\text{Time}(M) < \text{Time}(CS)$
- $\text{Alldiff}(T_x, R_x)$

(iii) Explain **ONE (1)** real-life example of CSP in the manufacturing field. (2 marks)

- In a car factory, tasks such as welding, painting, and packaging need to be assigned to different workstations.
- Each task must follow the correct sequence, stay within the time limit, and use the available resources or machines.
- Helps the factory find a suitable assignment of tasks to workstations while satisfying all these constraints.